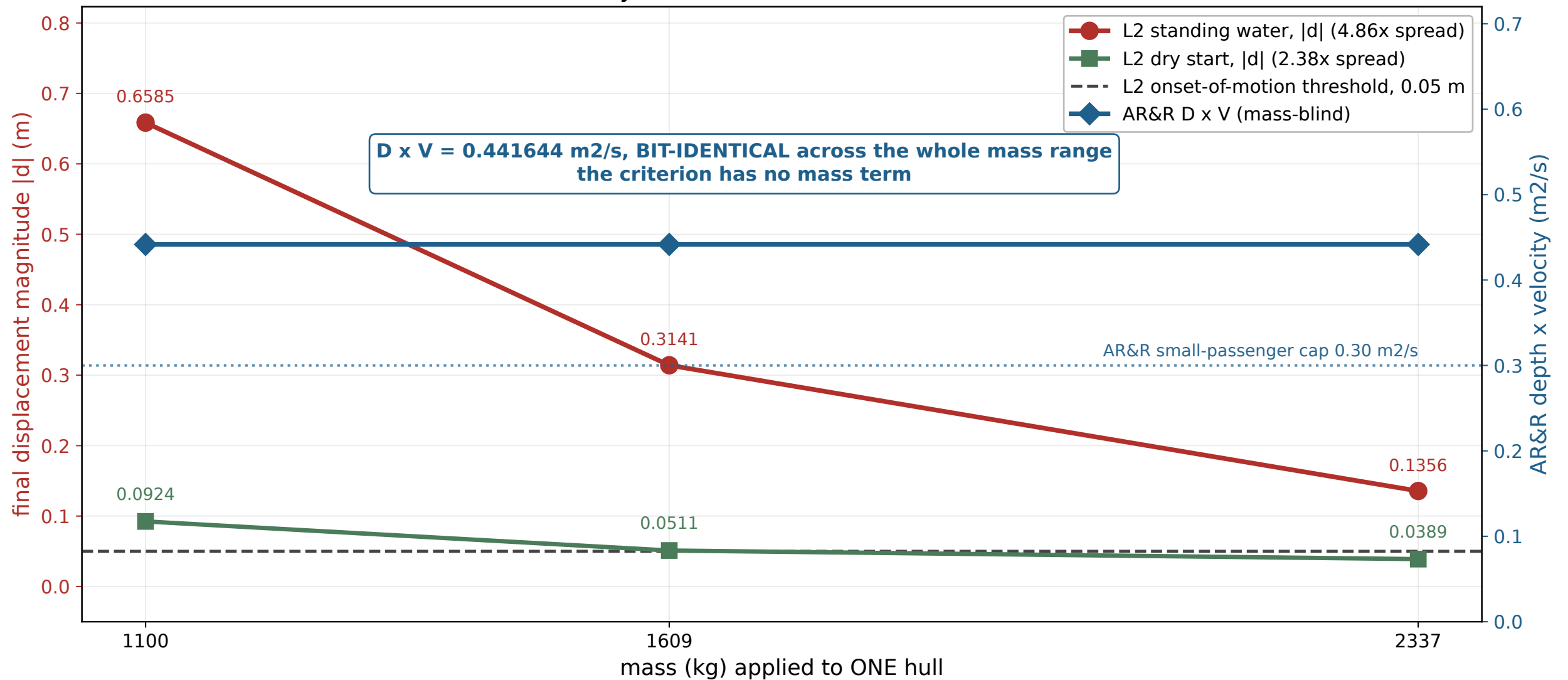


G5 Mass sensitivity on one hull: L2 resolves what D x V cannot



SCOPE, NON-NEGOTIABLE: all six runs plotted here use the SAME mesh, vehicle_geometry_research/yaris_coarse_v1l_watertight.ply, at 8905 solid particles (standing) and 8904 to 8905 (dry start). This is THREE MASSES ON ONE HULL, a mass sensitivity study. It is NOT three vehicles, and the word "class" may not appear unqualified for these rows. Where a class name is used elsewhere it names only which AR&R limit set was applied by kerb weight.

Left axis: the coupled simulation resolves a 4.86x displacement spread in standing water (0.658537 to 0.135559 m) and 2.38x in dry start (0.092399 to 0.038901 m). Right axis: the AR&R product is bit-identical at 0.441644 m2/s across the same 2.1x mass range, because D x V contains no mass term at all.

The two scenarios are not a controlled pair (different drivers, sustained inflow in one only), so read each series against itself, not against the other. Grid 64, realized depth 0.2944 m, 1.5 m/s surge, all runs deterministic.

Source: data/all_runs_inventory.csv and renders/yaris_render_s1/gates_results.json.